



~6,000 people · 1 laptop · 1 tablet · power fails weekly

## UNIT 1 · FOUNDATIONS AND PROCESS

CSC 4305 · Session 02 of 28

# SANTE-BASSAM: The Project

One project for the whole semester — the health post, what the system is and is not, the six questions every team must answer before writing code, and how a four-student team works.

### DATE

**Friday, 11 September 2026**

### MEETING TIME

**10:00 - 11:20 AM**

### DURATION

**80 minutes**

**5** concepts    **1** worked on SANTE-BASSAM    **1** assigned to you    **2** figures

## 0 The session in one picture

One project for the whole semester — the health post, what the system is and is not, the six questions every team must answer before writing code, and how a four-student team works.

3.1

### Why one project for the whole semester

Session 1 argued that engineering is what happens before the treatment and what makes the treatment accountable afterwards.

3.2

### The health post

A community health post in the Grand-Bassam district serves a catchment of roughly 6,000 people across a small town and 23 surrounding villages.

3.3

### What SANTE-BASSAM is — and is not

SANTE-BASSAM is a patient registry and appointment system : software that replaces the register book, the card index and the appointment book, runs on the two devices the post already has, keeps worki

3.4

### The six questions, answered for SANTE-BASSAM

Session 1 ended with six questions a team should be able to answer in writing before its first line of code.

3.5

### How the team works

Practice Agreement Team  
Four students; roles rotate per phase (a Product Owner proxy, a Scrum Master, two others — Session 5 defines these).

## 1 Learning outcomes

**L0 2.1** Describe the community health post SANTE-BASSAM serves — its people, its paper artifacts, its daily workflow, and its non-negotiable constraints — without reference to any software.

**L0 2.2** State what SANTE-BASSAM is and is not: the capabilities in scope this semester, the ones explicitly excluded, and why the line is drawn there.

**L0 2.3** Answer the six questions from Session 1 for SANTE-BASSAM, with a named person, a number, and a dated decision where each demands one.

**L0 2.4** Name the six delivery phases of the semester, the milestone artifact each produces, and which sessions build it.

LO 2.5

Explain how a decision made in one phase constrains the next, using at least two concrete SANTE-BASSAM examples.

LO 2.6

Set up the team's working agreement — roles, repository, meeting rhythm, the fortnightly visit to the post.



**How this session is built.** Worked in full on SANTE-BASSAM: The six questions, answered for SANTE-BASSAM. **Defined but left for you:** The six questions, answered for SANTE-BASSAM — the exercise or this week's homework.

## 2 Where the project stands

### THE SYSTEM

SANTE-BASSAM — a patient registry and appointment system for a community health post serving about 6,000 people in the Grand-Bassam district.

### THE PEOPLE

A front-desk clerk, a nurse-in-charge, two community health workers (CHWs) who run village outreach, a district health officer who needs monthly reports, and patients or their guardians.

### THE CONSTRAINTS

Electricity fails roughly weekly. Connectivity is intermittent. Hardware is one front-desk laptop and one Android tablet. Health records are governed by law and must use the national health identifier format.

### TODAY

A paper register, a card per patient, an appointment book, and the nurse's private notebook of follow-ups.

### YOUR TEAM

Four students, fourteen weeks, no budget, one instructor standing in for the health post when a real stakeholder is not available.

## 3.1 Why one project for the whole semester

Session 1 argued that engineering is what happens before the treatment and what makes the treatment accountable afterwards. You cannot learn that from exercises that start fresh each week, because the discipline lives in the **consequences**: the requirement you wrote in

week 3 is the test you cannot pass in week 12; the interface you fixed in week 5 is the reason week 9 goes smoothly. So this course uses one system, carried from first conversation to hand-over, and every session is applied to it.

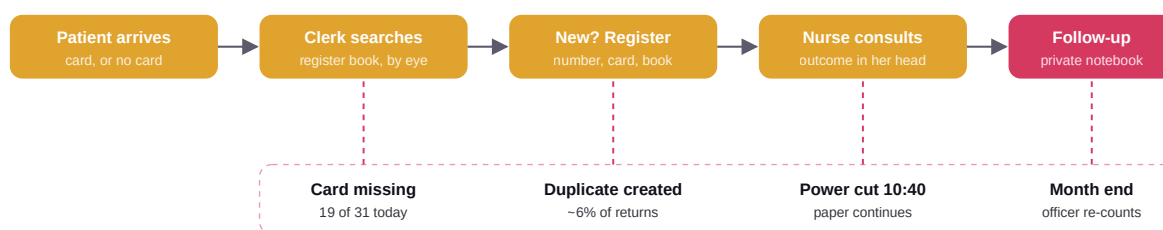
The system had to satisfy three conditions. It had to be **real enough to be constrained** — real users, real law, real power failures — because unconstrained problems teach nothing. It had to be **small enough for four students in fourteen weeks**. And it had to be **close enough to visit**, because the single most valuable engineering activity in this course is sitting beside the person who will use what you build.

## 3.2 The health post

A community health post in the Grand-Bassam district serves a catchment of roughly 6,000 people across a small town and 23 surrounding villages. It is staffed by a **nurse-in-charge**, who runs consultations and everything else; a **front-desk clerk**, who registers patients, finds their records and books them in; two **community health workers** (CHWs), who run outreach sessions in the villages several days a week; and a **pharmacy assistant**. A **district health officer** visits monthly and takes away an attendance report that feeds the national statistics. A **management committee** of local notables meets quarterly and can stop anything.

The post's record system is paper, and it works. A **register book** lists every patient ever registered, in order of registration, with a number. A **patient card** carries that number and is kept by the patient. An **appointment book** lists who is expected on which day. The nurse keeps a **private notebook** of patients she wants to see again, because the register cannot tell her that. At month end the district officer **re-counts the register by hand** to fill the Ministry's template.

### ONE THURSDAY MORNING AT THE FRONT DESK



Top row: the workflow as the post describes it. Bottom row: what one Thursday morning actually contained. The bottom row is the requirement.

The top row is what the post will tell you in an interview. The bottom row is what you see if you sit there for three hours. Session 7 is about the difference.

Three facts about the environment are not going to change and you did not choose them.

**Electricity fails roughly once a week**, usually for an hour, sometimes for a day.

**Connectivity is intermittent**: the town has mobile data most of the time; the villages have it some of the time; the post's own connection drops when the power does. And the hardware is **one laptop at the front desk and one Android tablet** for outreach, both donated, neither new.



**The first engineering decision has already been made for you.**

Anything that needs the network to work will not be used. Anything that loses a record when the power fails will be abandoned within a week and the post will return to paper. Every later decision in this course is downstream of these two sentences.

### 3.3

## What SANTE-BASSAM is — and is not

SANTE-BASSAM is a **patient registry and appointment system**: software that replaces the register book, the card index and the appointment book, runs on the two devices the post already has, keeps working when the power and the network do not, and produces the monthly report without a re-count. That is the whole ambition. It is enough.

#### THE SYSTEM

SANTE-BASSAM — a patient registry and appointment system for a community health post serving about 6,000 people in the Grand-Bassam district.

#### THE PEOPLE

A front-desk clerk, a nurse-in-charge, two community health workers (CHWs) who run village outreach, a district health officer who needs monthly reports, and patients or their guardians.

#### THE CONSTRAINTS

Electricity fails roughly weekly. Connectivity is intermittent. Hardware is one front-desk laptop and one Android tablet. Health records are governed by law and must use the national health identifier format.

#### TODAY

A paper register, a card per patient, an appointment book, and the nurse's private notebook of follow-ups.

#### YOUR TEAM

Four students, fourteen weeks, no budget, one instructor standing in for the health post when a real stakeholder is not available.

## In scope this semester

### REGISTRY

Register a patient; find a returning patient — including one with no card; detect probable duplicates

### APPOINTMENTS AND VISITS

Book and reschedule; record each visit as attended, missed or rescheduled; the nurse's follow-up list

### OUTREACH

Register and record on the tablet with no network; sync to the desk when connectivity returns

### REPORTING

The district officer's monthly report by village and outcome, in the Ministry template, exportable

## Explicitly out of scope

### CLINICAL NOTES

What the nurse writes about a consultation stays on paper. A clinical record is a different system with different law

### PHARMACY STOCK

The assistant reads visit outcomes; the system does not manage inventory

### TIME-SLOT SCHEDULING

The post books a day, not an hour. So does the system

### MULTIPLE POSTS

One post, one server. Records carry a post identifier so that a district system could merge them later — and that is all

The out-of-scope list is as important as the in-scope one. In week 10 someone — a stakeholder, or one of you — will propose adding clinical notes because it seems small. The answer is written here, dated today, and the reason is not effort: it is that a clinical record changes who may read the data, how long it is kept, and what a defect costs. Scope is a requirement.

## 3.4 The six questions, answered for SANTE-BASSAM

Session 1 ended with six questions a team should be able to answer in writing before its first line of code. Here are five of them answered — with the specificity the course will hold you to. The sixth is left for you.

### WORKED

## Q1 — Who is this for, what do they do today, and what will they stop doing?

**A person, not a category.** The primary user is the front-desk clerk — call her Madame Koné, because you will meet her. Today she scans the register book by eye to find returning

patients, writes each new patient three times (book, card, appointment book), and re-registers about one returning patient in twenty because she could not find them. She will *stop* scanning the book, stop writing the third copy, and stop guessing whether a mother's child is already registered.

The secondary users are the nurse (who will stop keeping the notebook), the CHWs (who will stop bringing paper sheets back from the village for the clerk to transcribe), and the district officer (who will stop re-counting). Each of them gives up a habit, and each habit is a workaround that encodes a requirement. Session 7 shows you how to read them.

#### WORKED

### Q2 — What must be true for this to be worth building?

**A number someone can judge you by in six months.** Three of them, measured now and again in December:

| MEASURE                                                       | TODAY<br>(PAPER)                  | WORTH BUILDING<br>IF...                                    | HOW<br>MEASURED                    |
|---------------------------------------------------------------|-----------------------------------|------------------------------------------------------------|------------------------------------|
| <b>Time to find a returning patient with no card</b>          | ~45 s<br>median; fails<br>~1 in 5 | $\leq 60$ s, fails $\leq 1$ in 10<br>(AT-01, Session 3)    | Stopwatch at the desk, 30 patients |
| <b>Duplicate registrations per 100 returning visits</b>       | ~6                                | $\leq 1$                                                   | Count in one month of records      |
| <b>District officer's hours to produce the monthly report</b> | ~3 h re-counting                  | $\leq 10$ min, and she submits without re-counting (AT-03) | She tells you; then you watch      |

If in December the first number is 90 seconds, the project failed even if every test passed. Write these three in the Project Plan and do not soften them later.

## WORKED

## Q3 — Which constraints are not negotiable?

| CONSTRAINT                                                                           | WHY IT IS FIXED                           | WHAT IT FORCES                                                                                                                |
|--------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Power fails weekly</b>                                                            | Infrastructure; not yours to change       | Every save is durable before the confirmation appears; the tablet on battery is the fallback for the desk                     |
| <b>Network is intermittent</b>                                                       | Geography and budget                      | Offline-first: nothing the clerk or CHW does during a clinic day may depend on the server                                     |
| <b>One laptop, one tablet, both donated</b>                                          | No budget                                 | Performance targets are measured on <i>these</i> devices; nothing heavier than they can run                                   |
| <b>Health-record law and the national identifier format</b>                          | Statute                                   | Identifier format, consent for minors, retention and void-not-delete are domain requirements, never backlog items (Session 9) |
| <b>The Ministry report template</b>                                                  | The officer will not accept anything else | Report columns and order are fixed; only the counting is ours                                                                 |
| <b>French at the desk; some clerks and CHWs more comfortable reading than typing</b> | The people                                | Interface in French; fields in the order of the paper card; typing minimised (village list, not free text)                    |
| <b>Fourteen weeks, four students, other courses</b>                                  | The calendar                              | Scope is fixed by 2.3; what flexes is polish, never durability                                                                |

## WORKED

## Q4 — Which decision, if wrong, is most expensive to reverse?

**That is your architecture. Write it down with its reason, and date it.** For SANTE-BASSAM the answer is *where the data lives*. Two candidates:

## A · Server-centred

### SHAPE

One database on a server; laptop and tablet are clients that need a connection

### COST IF WRONG

Discovered in week 8 when the CHW cannot register in a village; the whole tablet path is rebuilt; ~40 team-days

## B · Offline-first (chosen, 11 Sep 2026)

### SHAPE

Each device has its own complete local store; the server is only a sync target; records carry locally generated identifiers

### COST IF WRONG

If sync proves unreliable, the desk still works alone. Rebuild limited to the sync module; ~10 team-days

B is chosen not because it is easier — it is harder — but because the cost of being wrong is a quarter of A's, and because Q3 makes A's failure near-certain. Session 3 will show you how to *test* this decision before you commit fourteen weeks to it; the decision record itself goes in the repository today.

## WORKED

## Q5 — What is the single point of failure?

| IF THIS ONE THING IS UNAVAILABLE...         | ...WHAT HAPPENS                                                           | MITIGATION THE DESIGN MUST INCLUDE                                                                                               |
|---------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>The laptop (stolen, broken, dropped)</b> | Every record since the last sync is gone; the post cannot register anyone | Sync to the server whenever there is a connection; a weekly export to a USB key the nurse keeps; the tablet can run the desk app |
| <b>The tablet, in a village</b>             | Outreach records since the last sync are gone                             | Sync before leaving the post and on return; a paper fallback sheet in the CHW's bag (this is Q6)                                 |
| <b>The clerk (sick, on leave)</b>           | Nobody knows how to use the system                                        | The nurse is trained too; a one-page sheet at the desk; no step needs a password only the clerk knows                            |
| <b>Your team, after December</b>            | Nobody can fix anything                                                   | The hand-over sheet, self-explanatory errors, automatic backup — Session 28 onward                                               |

| IF THIS ONE THING IS UNAVAILABLE... | ...WHAT HAPPENS               | MITIGATION THE DESIGN MUST INCLUDE                    |
|-------------------------------------|-------------------------------|-------------------------------------------------------|
| <b>The server</b>                   | Sync stops; nothing else does | By design (Q4). The post can run without it for weeks |

Notice the last row. Q4 was answered so that the server is *not* a single point of failure. That is what it means for one decision to constrain the next.

ASSIGNED — NOT WORKED

### Q6 — What is the fallback when the clever thing fails?

Denver's airport had none. Your registry will run in a building where the power fails weekly and a village where the network does not exist. Q1–Q5 above give you the pattern: a named person, a number, a table with a mitigation column. This week you answer Q6 for SANTE-BASSAM to the same standard — for the desk, for outreach, and for the monthly report. The brief is in section 4.

## 3.5 How the team works

| PRACTICE          | AGREEMENT                                                                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Team</b>       | Four students; roles rotate per phase (a Product Owner proxy, a Scrum Master, two others — Session 5 defines these). One repository per team on the course GitHub organisation                                          |
| <b>Rhythm</b>     | Two-week sprints from week 5 (Session 5). Before that, one spike to retire the largest risk (Session 3)                                                                                                                 |
| <b>The post</b>   | A fortnightly visit — one review with the nurse, one morning at the desk. The instructor stands in for the post when a visit cannot happen; the substitution is recorded                                                |
| <b>Artifacts</b>  | Everything versioned: <code>plan/</code> , <code>srs/</code> , <code>arch/</code> , <code>src/</code> , <code>tests/</code> , <code>handover/</code> . A decision without a dated record in the repository was not made |
| <b>Milestones</b> | Six, at the ends of the phases above; each is a tagged commit reviewed by the instructor and, for P1 and P5, by the nurse                                                                                               |

**Thursday morning, replayed (pairs, 15 minutes).** Using the figure in 2.2 and the scenario below, work out what the system must do at each moment — and what it must not.

09:10. A mother arrives with a two-year-old and no card. She gives the child's first name and her own. 09:14. The clerk finds two children of that name with mothers of that name, in different villages. 09:30. A neighbour brings a child whose mother is away. 10:40. The power fails while the clerk is halfway through registering a newborn. 11:05. The nurse tells the clerk to “put Awa on the Monday list”. 11:30. A CHW returns from a village with eleven registrations on paper.

- 1 For each moment, name the actor, the paper artifact involved today, and the single question the system must answer at that moment.
- 2 Mark the two moments where a wrong answer is most expensive — and say what “expensive” means for each (lost record? wrong patient? legal?).
- 3 Identify one moment that is *out of scope* per 2.3, and write the one-sentence reply you would give a stakeholder who asks for it anyway.
- 4 Which of the six phases in 2.5 will first produce an artifact that addresses the 10:40 moment? Name the artifact.



**Discussion points to expect.** The 09:14 moment is the whole search problem: two candidates, disambiguated by village and birth year — that is a ranking requirement, not a lookup. The 09:30 moment is the consent question (a domain requirement). Most pairs put 10:40 in P4 (testing); it is first addressed in P0, today, by Q4, and tested in P4. The phase that *decides* and the phase that *proves* are different, and both matter.

Due before Session 3 (Tuesday 15 September), pushed to your team repository under

plan/

- ☐ **Q6, answered (the unworked question).** `plan/q6-fallbacks.md`. For each of the front desk, outreach, and the monthly report: what is the clever thing, what is the specific way it fails (name the failure, not “an error”), what the person does in the next five minutes, and how the system recovers when the clever thing returns — including how paper records taken during the outage get in. Use the table form of Q5, and give at least one number (how long the fallback must cover: an hour? a day? a week?).
- ☐ **The six questions, as a team document.** `plan/six-questions.md`: Q1-Q5 in your own words — you may disagree with this document, with reasons — plus your Q6. Names attached, dated.
- ☐ **Team charter.** `plan/charter.md`: members, the rotation of roles, meeting times, the day of the fortnightly visit, and the rule for what happens when someone misses a commitment.
- ☐ **Baseline measurements.** During your first visit (or from the instructor if the visit is not yet possible): the three numbers in Q2, measured or estimated with the method stated.
- ☐ **Reading.** Sommerville, Chapter 1, sections 1.1–1.3 (if not already done) and Chapter 2, section 2.1, before Session 3.



**Moved to reading.** The six-phase delivery map (P0 Foundation → P5 Delivery) is on the course site as the session calendar; the working-agreement template from the team-organisation reading is in `plan/working-agreement.md`.



**Assessment link.** These four files are the first section of the Project Plan, the P0 milestone due at the end of Session 5. The plan is the only artifact in the course that every later artifact must remain consistent with — write it as something you will be held to.

6

## One-page recap

### THE INSIGHTS TO CARRY FORWARD

#### THE HEALTH POST

**The first engineering decision has already been made for you.** Anything that needs the network to work will not be used. Anything that loses a record when the power fails will be abandoned within a week and the post will return to paper. Every later decision in this course is downstream of these two sentences.

## Key terms

community health post

register book

patient card

appointment book

workaround

catchment

scope (in / out)

non-negotiable constraint

offline-first

single point of failure

fallback

milestone artifact

delivery phase

decision record

## Readings and references

Sommerville, I. *Software Engineering*, 10th ed. — Chapter 1, sections 1.1-1.3; Chapter 2, section 2.1.

World Health Organization. *Analysing Disrupted Health Sectors*, Module 2 — health information at primary level (context reading, optional).

Brooks, F. *The Mythical Man-Month*, chapter 1 (“The Tar Pit”) — why a program and a programming systems product differ by a factor of nine.

**Pressman and Maxim**, Chapter 1 (The Nature of Software).

International University of Grand-Bassam · School of Science and Engineering · **CSC 4305**, Fall 2026

Next · Session 03 — Plan-Driven Process Models I: Waterfall and the V-model